Assignment 1

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Answer 1

$$A = \begin{bmatrix} 1.2 & 0.2 \\ 0.7 & 1.2 \end{bmatrix}$$

$$z[0] = \begin{bmatrix} 0.2 & 0.3 \\ 0.2 & 0.3 \end{bmatrix}$$

Answer 2

Method 1: Interval Matrix Method

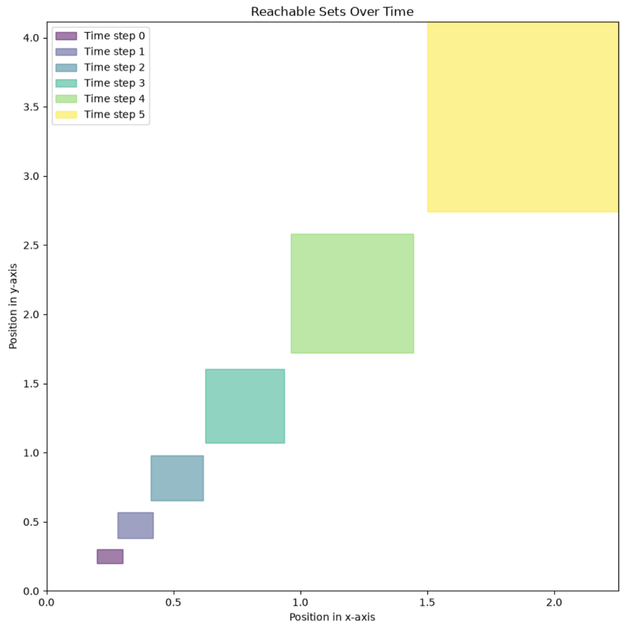

Figure 1: Reachable sets using the interval matrix method up to $T = 5$.

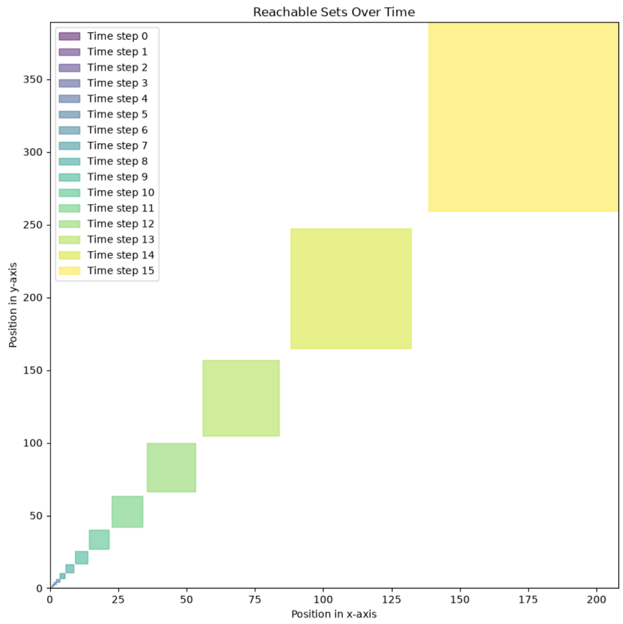

Figure 2: Reachable sets using the interval matrix method up to $T = 15$.

Method 2: Vertex Shooting Method

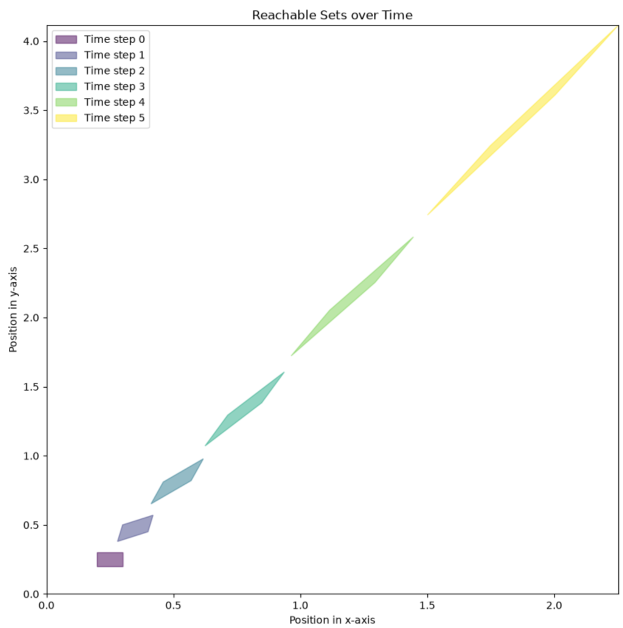

Figure 3: Reachable sets using the vertex shooting method up to $T = 5$.

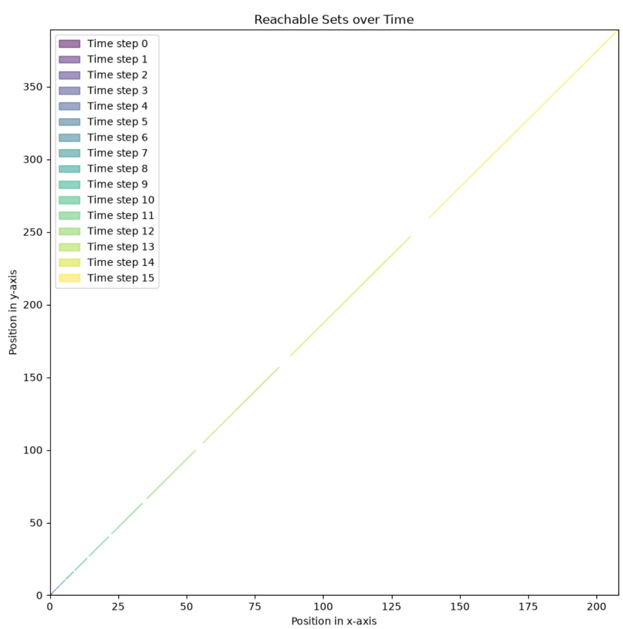

Figure 4: Reachable sets using the vertex shooting method up to $T = 15$.

Answer 3

Trajectories from five random points from the initial set are plotted alongside the reachable sets computed using the vertex shooting method.

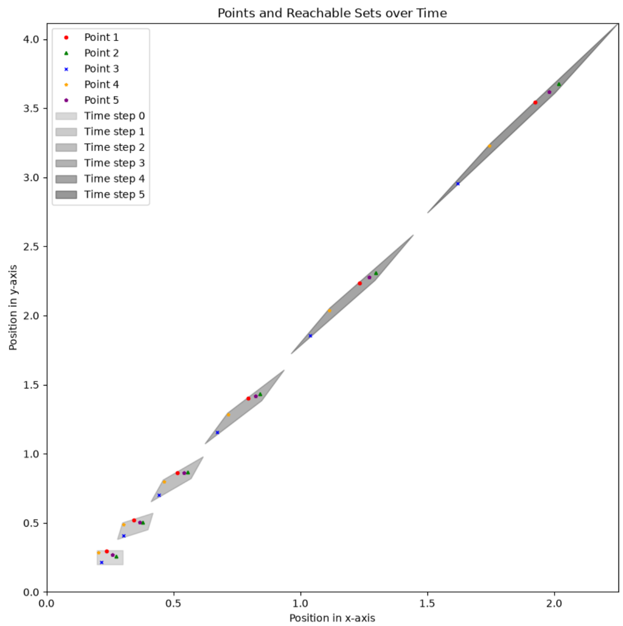

Figure 5: Random point trajectories and their corresponding reachable sets up to $T = 5$.

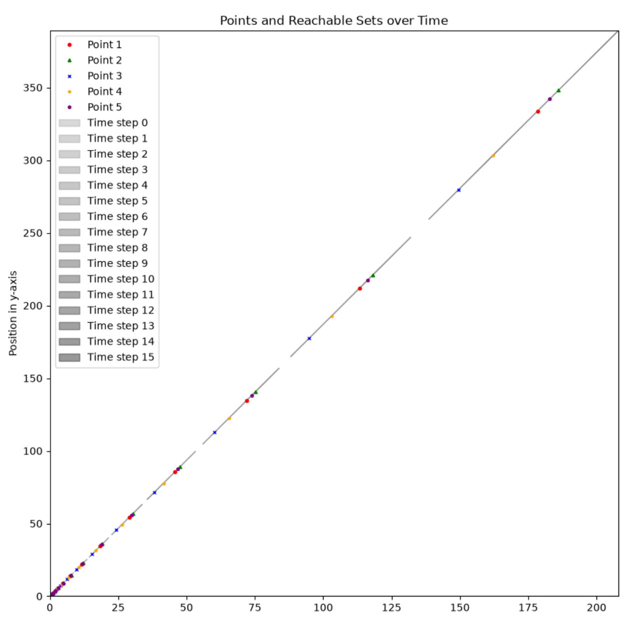

Figure 6: Random point trajectories and their corresponding reachable sets up to $T = 15$.